

Qualification Pack



RPA Developer

QP Code: SSC/Q8604

Version: 3.0

NSQF Level: 5.5

IT-ITeS SSC NASSCOM || NASSCOM Plot No - 7, 8, 9 & 10, 3rd Floor, Sector 126
Noida Uttar Pradesh - 201303 || email:sscstandards@nasscom.in

Qualification Pack

Contents

SSC/Q8604: RPA Developer	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
SSC/N8601: Identify and analyse business processes suitable for automation	5
SSC/N8603: Establish a functional design of the automation solution	9
SSC/N8607: Define infrastructure requirements for the Robotic Process Automation solution	14
SSC/N8614: Design & Develop the bot as per design documents	18
SSC/N8615: Develop test cases for the bot	24
SSC/N8616: Identify and fix bugs in the bot	28
SSC/N8618: Ensure that processes are being triggered as intended and handle process exceptions ...	32
SSC/N8622: Review post deployment analytics & provide process improvement information to appropriate stakeholders	37
SSC/N8629: New Technology application in RPA	42
DGT/VSQ/N0102: Employability Skills (60 Hours)	46
Assessment Guidelines and Weightage	53
<i>Assessment Guidelines</i>	53
<i>Assessment Weightage</i>	53
Acronyms	55
Glossary	56

Qualification Pack

SSC/Q8604: RPA Developer

Brief Job Description

This qualification trains individuals to become RPA Developers, covering the complete RPA lifecycle. It begins with foundational knowledge of RPA, business process analysis, and bot design principles, progressing to hands-on development using various tools and programming languages. The curriculum emphasizes testing, deployment, and monitoring techniques, including security and infrastructure considerations. Advanced topics such as AI integration and cloud-based solutions are also included. Finally, the qualification integrates essential employability skills training to prepare graduates for the workforce.

Personal Attributes

They must possess strong interpersonal skills to coordinate with other team members and achieve project success. They should also have a basic understanding of various technologies.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SSC/N8601: Identify and analyse business processes suitable for automation](#)
2. [SSC/N8603: Establish a functional design of the automation solution](#)
3. [SSC/N8607: Define infrastructure requirements for the Robotic Process Automation solution](#)
4. [SSC/N8614: Design & Develop the bot as per design documents](#)
5. [SSC/N8615: Develop test cases for the bot](#)
6. [SSC/N8616: Identify and fix bugs in the bot](#)
7. [SSC/N8618: Ensure that processes are being triggered as intended and handle process exceptions](#)
8. [SSC/N8622: Review post deployment analytics & provide process improvement information to appropriate stakeholders](#)
9. [SSC/N8629: New Technology application in RPA](#)
10. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	IT-ITeS
--------	---------

Qualification Pack

Sub-Sector	Future Skills
Occupation	Robotic Process Automation
Country	India
NSQF Level	5.5
Credits	18
Aligned to NCO/ISCO/ISIC Code	NCO-2015/2512.NIL
Minimum Educational Qualification & Experience	Completed 3 year UG degree (of 3-year/ 4-year UG.PG/UG/Degree/Diploma in (Engg./Science)) OR Completed 3-year diploma (after 10th) (UG Diploma*after 10th .PG/UG/Degree/Diploma in (Engg./Science)) with 3 Years of experience in relevant field.Relevant experience in job roles related to RPA/Computer Science/IT The relevant experience would include work, internship and apprenticeship OR Previous relevant Qualification of NSQF Level (5) with 1.5 years of experience in relevant filed.Relevant experience in job roles related to RPA/Computer Science/IT The relevant experience would include work, internship and apprenticeship
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	18/02/2028
NSQC Approval Date	18/02/2025
Version	3.0
Reference code on NQR	QG-5.5-IT-03656-2025-V2-NASSCOM
NQR Version	2

Qualification Pack

SSC/N8601: Identify and analyse business processes suitable for automation

Description

This unit is about analyzing business processes and identifying processes suitable for automation

Scope

The scope covers the following :

- RPA Introduction
- Process Mapping
- RPA Suitability Assessment
- Value Identification & ROI Calculation
- Stakeholder Analysis

Elements and Performance Criteria

RPA Introduction

To be competent, the user/individual on the job must be able to:

PC1. Understand the RPA, Intelligent automation ,hyperautomation and Agentic AI

PC2. Understand the RPA Lifecycle and its use cases

Process Mapping

To be competent, the user/individual on the job must be able to:

PC3. Map a real-world business process using various techniques (flowcharts, BPMN)

RPA Suitability Assessment

To be competent, the user/individual on the job must be able to:

PC4. Evaluate the mapped process against the criteria for RPA suitability

Value Identification & ROI Calculation

To be competent, the user/individual on the job must be able to:

PC5. Estimate the potential ROI (cost estimations,time savings, error reduction) from automation

Stakeholder Analysis

To be competent, the user/individual on the job must be able to:

PC6. Identify key stakeholders impacted by the automation of the chosen process

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Organizational policies, procedures and guidelines which relate to collecting and defining business requirements
- KU2.** Different data sources and how to access documents and information from data sources
- KU3.** Organizational policies and procedures for sharing data

Qualification Pack

- KU4.** Organizational policies and procedures for documenting requirements and creating IPA
- KU5.** Who to involve when defining requirements and identifying the processes to be automated
- KU6.** Standard templates and tools available and how to use them
- KU7.** Understanding of various use cases of Robotic Process Automation across different verticals
- KU8.** Identification of automation potential of processes
- KU9.** Important KPIs pertaining to Robotic Process Automation
- KU10.** Evaluate and define requirements

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Seek clarification and advice from appropriate people
- GS2.** Conduct business contextual discussions
- GS3.** Listen effectively and orally communicate information accurately
- GS4.** Work effectively in a customer facing environment
- GS5.** Create accurate and well written documents/reports

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>RPA Introduction</i>	10	-	-	4
PC1. Understand the RPA, Intelligent automation ,hyperautomation and Agentic AI	5	-	-	2
PC2. Understand the RPA Lifecycle and its use cases	5	-	-	2
<i>Process Mapping</i>	5	13	-	4
PC3. Map a real-world business process using various techniques (flowcharts, BPMN)	5	13	-	4
<i>RPA Suitability Assessment</i>	5	13	-	4
PC4. Evaluate the mapped process against the criteria for RPA suitability	5	13	-	4
<i>Value Identification & ROI Calculation</i>	5	12	-	4
PC5. Estimate the potential ROI (cost estimations,time savings, error reduction) from automation	5	12	-	4
<i>Stakeholder Analysis</i>	5	12	-	4
PC6. Identify key stakeholders impacted by the automation of the chosen process	5	12	-	4
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8601
NOS Name	Identify and analyse business processes suitable for automation
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Robotic Process Automation
NSQF Level	5.5
Credits	1
Version	3.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

SSC/N8603: Establish a functional design of the automation solution

Description

This unit is about defining a functional design of the automation solution by refining business processes, creating Process Design Document and defining other Robotic Process Automation milestone deliverables

Scope

The scope covers the following :

- Reengineering Processes
- Creating Process Design Document
- Creating other Robotic Process Automation Milestone Deliverables

Elements and Performance Criteria

Reengineering processes

To be competent, the user/individual on the job must be able to:

- PC1.** Break down the selected business process(es) to granular level of detail
- PC2.** Re-engineer the identified process to improve automation potential using design thinking and other frameworks

Creating Process Definition Document

To be competent, the user/individual on the job must be able to:

- PC3.** Agree with the development team on the level of detail needed by them
- PC4.** Conduct workshops and partner with SMEs to create process flows and process maps of 'as-is' and 'to-be' processes
- PC5.** Define if the bot in a given process is attended or unattended
- PC6.** Prepare the Process Definition/Design Document (PDD) containing essential elements of 'as-is' process description, 'to-be' process description and exceptions using appropriate tool
- PC7.** Create a checklist to ensure the documentation is accurate and consistent
- PC8.** Get the Process Design Document validated by various business and technical stakeholders

Creating Other Robotic Process Automation milestone deliverables

To be competent, the user/individual on the job must be able to:

- PC9.** Define Business Continuity Plan (BCP), Non functional requirements (NFR) for the whole RPA program and functional requirement questionnaire (FRQ)
- PC10.** Coordinate the review and submission of RPA milestone deliverables (i.e. Solution Design Document (SDD), Operational Impact Document (OID), Process Design Instruction (PDI), Object Design Instructions (ODI), Technical Design Document (TDD))
- PC11.** Maintain automation documentation according to standards, best practices and standard operating procedures

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

Qualification Pack

- KU1.** Organizational policies, procedures and guidelines which relate to automation of processes
- KU2.** Different data sources and how to access documents and information from data sources
- KU3.** Organizational policies and procedures for documenting Process Design Document
- KU4.** Who to involve while defining functional and non-functional requirements, and defining and validating Process Design Document
- KU5.** The range of standard templates and tools available and how to use them
- KU6.** Business contextual discussions
- KU7.** Optimization of automation potential of processes
- KU8.** Processes and workflows pertaining to the Robotic Process Automation solution
- KU9.** Attended and unattended bots features and uses
- KU10.** Frameworks to ensure business continuity
- KU11.** Functional and non-functional requirements and their usage in different processes
- KU12.** Best practices in preparing Process Design Documents
- KU13.** Other Robotic Process Automation milestone deliverables (e.g., Solution Design Document, Technical Design Document, Object Design Instructions, etc.)

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** seek clarification and advice from appropriate people
- GS2.** Listen effectively and orally communicate information accurately
- GS3.** Follow rule-based decision-making processes
- GS4.** Make decisions on suitable courses
- GS5.** Plan and organize the work to achieve targets and deadlines
- GS6.** Problem-solving approaches in different situations
- GS7.** Balanced judgments for different situations
- GS8.** Accurate and well written documents/reports

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Reengineering processes</i>	8	10	-	4
PC1. Break down the selected business process(es) to granular level of detail	4	5	-	2
PC2. Re-engineer the identified process to improve automation potential using design thinking and other frameworks	4	5	-	2
<i>Creating Process Definition Document</i>	15	27	-	12
PC3. Agree with the development team on the level of detail needed by them	2	4	-	2
PC4. Conduct workshops and partner with SMEs to create process flows and process maps of 'as-is' and 'to-be' processes	2	4	-	2
PC5. Define if the bot in a given process is attended or unattended	3	5	-	2
PC6. Prepare the Process Definition/Design Document (PDD) containing essential elements of 'as-is' process description, 'to-be' process description and exceptions using appropriate tool	3	5	-	2
PC7. Create a checklist to ensure the documentation is accurate and consistent	3	5	-	2
PC8. Get the Process Design Document validated by various business and technical stakeholders	2	4	-	2
<i>Creating Other Robotic Process Automation milestone deliverables</i>	7	13	-	4
PC9. Define Business Continuity Plan (BCP), Non functional requirements (NFR) for the whole RPA program and functional requirement questionnaire (FRQ)	3	5	-	2
PC10. Coordinate the review and submission of RPA milestone deliverables (i.e. Solution Design Document (SDD), Operational Impact Document (OID), Process Design Instruction (PDI), Object Design Instructions (ODI), Technical Design Document (TDD))	2	4	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Maintain automation documentation according to standards, best practices and standard operating procedures	2	4	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8603
NOS Name	Establish a functional design of the automation solution
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Robotic Process Automation
NSQF Level	5.5
Credits	2
Version	3.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

SSC/N8607: Define infrastructure requirements for the Robotic Process Automation solution

Description

This unit is about defining the infrastructure requirements for the Robotic Process Automation solution

Scope

The scope covers the following :

- Infrastructure Requirements
- Security Requirements
- Other Requirements
- Infrastructure Maintenance
- Business Continuity

Elements and Performance Criteria

Infrastructure Requirements

To be competent, the user/individual on the job must be able to:

- PC1.** Identify suitable infrastructure in network, storage, and compute, such as on-premise storage/cloud/hybrid
- PC2.** Define the environments and corresponding runtime resource
- PC3.** Gather specifications on databases, installation modes, servers, control room architecture and communication handshake between bot agent and control room from appropriate teams
- PC4.** Evaluate if the solution is non-invasive on existing infrastructure landscape and network/security architecture

Security Requirements

To be competent, the user/individual on the job must be able to:

- PC5.** Specify the security requirement encompassing permission, user access, protocols, provisioning of bot IDs, role-based access controls etc, for both unattended and monitored bots

Other Requirements

To be competent, the user/individual on the job must be able to:

- PC6.** Define other prerequisites such as licenses, update/upgrade policies, etc

Infrastructure Maintenance

To be competent, the user/individual on the job must be able to:

- PC7.** Define downtime schedule and activities during downtime such as restart of runtime resources, clearing log files, check space etc.

Business Continuity

To be competent, the user/individual on the job must be able to:

- PC8.** Prepare a plan for high availability and disaster recovery

Qualification Pack

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Organizational policies, procedures and guidelines which relate to defining infrastructure requirements for the solution
- KU2.** Organizational policies and procedures for shortlisting third-party service providers for network, storage, security and other infrastructure aspects of the RPA solution, including license requirements
- KU3.** Organizational policies and procedures for documenting various standards, technical specifications and guidelines
- KU4.** Who to involve while defining and reviewing technical architecture of Robotic Process Automation ecosystem
- KU5.** The range of standard templates and tools
- KU6.** Organizational policies and procedures for repair, maintenance, or upkeep activities
- KU7.** Best practices for designing technical architecture and defining security requirements
- KU8.** Organizational policies and frameworks for business continuity
- KU9.** Understanding of processes and workflows pertaining to the Robotic Process Automation solution
- KU10.** Understanding of databases, servers, networking components, and other such technical infrastructure components
- KU11.** Various third party cloud storage, database and compute technologies
- KU12.** Understanding of security parameters (e.g., user access, bot ID provisioning, etc.), and how to define them

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Follow instructions, guidelines, procedures, rules and service level agreements
- GS2.** Contribute to the quality of team work
- GS3.** Work independently as well as collaboratively
- GS4.** accurate and well written documents/reports
- GS5.** Understand business impact of the Robotic Process Automation solution and disseminate relevant information to others
- GS6.** Seek, receive and act on gathered feedback from various sources
- GS7.** Share the assimilated data and feedback with stakeholders
- GS8.** Ensure work is complete and free from errors
- GS9.** Balanced judgments to different situations
- GS10.** Provide opinions on work in a detailed and constructive way
- GS11.** Make decisions on suitable courses
- GS12.** Plan and organize the work to achieve targets and deadlines
- GS13.** Problem-solving approaches in different situations

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Infrastructure Requirements</i>	14	25	-	10
PC1. Identify suitable infrastructure in network, storage, and compute, such as on-premise storage/cloud/hybrid	3	6	-	2
PC2. Define the environments and corresponding runtime resource	3	6	-	2
PC3. Gather specifications on databases, installation modes, servers, control room architecture and communication handshake between bot agent and control room from appropriate teams	3	6	-	3
PC4. Evaluate if the solution is non-invasive on existing infrastructure landscape and network/security architecture	5	7	-	3
<i>Security Requirements</i>	3	6	-	2
PC5. Specify the security requirement encompassing permission, user access, protocols, provisioning of bot IDs, role-based access controls etc, for both unattended and monitored bots	3	6	-	2
<i>Other Requirements</i>	4	6	-	2
PC6. Define other prerequisites such as licenses, update/upgrade policies, etc	4	6	-	2
<i>Infrastructure Maintenance</i>	4	6	-	3
PC7. Define downtime schedule and activities during downtime such as restart of runtime resources, clearing log files, check space etc.	4	6	-	3
<i>Business Continuity</i>	5	7	-	3
PC8. Prepare a plan for high availability and disaster recovery	5	7	-	3
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8607
NOS Name	Define infrastructure requirements for the Robotic Process Automation solution
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Robotic Process Automation
NSQF Level	5.5
Credits	2
Version	3.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

SSC/N8614: Design & Develop the bot as per design documents

Description

This unit is about building bots using the tools defined in the design documents

Scope

The scope covers the following :

- Workflow assessment
- Development Design
- Reusable Components
- Develop bot
- Integrate
- Best Practices
- Documentation

Elements and Performance Criteria

Workflow assessment

To be competent, the user/individual on the job must be able to:

- PC1.** Identify the process and objects from the Solution Design Document pertaining to the Robotic Process Automation solution
- PC2.** Assess the workflows and how different components of the bot/ Robotic Process Automation solution interact with each other

Development Design

To be competent, the user/individual on the job must be able to:

- PC3.** Define the specifications for each component of the bot
- PC4.** Define the tools to be used in developing the bot
- PC5.** Plan reallocation of employees with concerned stakeholders so that they perform higher value functions

Reusable Components

To be competent, the user/individual on the job must be able to:

- PC6.** Identify the modules in the code that can be reused
- PC7.** Create object and process library to validate reusability

Develop bot

To be competent, the user/individual on the job must be able to:

- PC8.** Identify the objects for bot development from the Technical Design Document
- PC9.** Identify the tools for developing bot as per the design documents
- PC10.** Generate the software code or bots for each component of Robotic Process Automation solution

Integrate

To be competent, the user/individual on the job must be able to:

Qualification Pack

PC11. Integrate various modules of the bot or Robotic Process Automation solution

Best Practices

To be competent, the user/individual on the job must be able to:

PC12. Develop modular codes or programs so that they can be reused

PC13. Use various techniques, such as use of configuration file, to avoid hardcoding external settings (file paths, URLs, etc.)

PC14. Apply code quality principles and best practices, such as secure coding practices naming convention standards and compliance, zero usage of junk code, componentization etc. to maintain readability of the code

PC15. Align the code development process to Agile methodology principles to coordinate with cross-functional teams, and ensure timely delivery of the bots

PC16. Ensure compliance with internal control or audit requirements

Documentation

To be competent, the user/individual on the job must be able to:

PC17. Document the software code for consumption and feedback from relevant stakeholders, and include pertinent information such as screenshots, possible exceptions, error handling, etc.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Organizational policies, procedures and guidelines which relate to building and maintaining bots

KU2. Organizational policies and procedures for sharing data

KU3. Organizational policies and procedures for documenting and managing security configurations

KU4. Who to involve while defining continuous integration and continuous delivery practices

KU5. The range of standard templates and tools

KU6. Understanding of processes and workflows pertaining to the Robotic Process Automation solution

KU7. Types of dependencies in bot development

KU8. Attended and unattended bots and their development

KU9. Scripting / programming languages to develop the bot

KU10. Various operating systems such as Linux, Ubuntu, or Windows

KU11. Tools to develop bots

KU12. Secure APIs development for integrating the bot

KU13. Resilient bots creation

KU14. Tools to automate bot build and testing

KU15. Types of version control system

KU16. Types of SDLC environments (e.g., development, testing, production, etc.)

KU17. Types of security standards and protocols used in bot development

KU18. Types of security authentication measures and tools used in Robotic Process Automation

KU19. Best practices in coding

Qualification Pack

KU20. Application of statistical models and ML algorithms in the context of Robotic Process Automation solutions

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Follow instructions, guidelines, procedures, rules and service level agreements
- GS2.** Work independently as well as collaboratively
- GS3.** Communicate with others in writing
- GS4.** Understand the impact of RPA solution related decisions on business and organization
- GS5.** Seek, receive and act on gathered feedback from various sources
- GS6.** Share the assimilated data and feedback with stakeholders
- GS7.** Ensure the work is complete and free from errors
- GS8.** Balanced judgments to different situations
- GS9.** Make decisions on suitable courses
- GS10.** Plan and organize the work to achieve targets and deadlines
- GS11.** Problem-solving approaches in different situations
- GS12.** Seek clarification on problems from others

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Workflow assessment</i>	2	8	-	4
PC1. Identify the process and objects from the Solution Design Document pertaining to the Robotic Process Automation solution	1	4	-	2
PC2. Assess the workflows and how different components of the bot/ Robotic Process Automation solution interact with each other	1	4	-	2
<i>Development Design</i>	4	11	-	4
PC3. Define the specifications for each component of the bot	1	4	-	2
PC4. Define the tools to be used in developing the bot	1	4	-	1
PC5. Plan reallocation of employees with concerned stakeholders so that they perform higher value functions	2	3	-	1
<i>Reusable Components</i>	4	8	-	2
PC6. Identify the modules in the code that can be reused	2	4	-	1
PC7. Create object and process library to validate reusability	2	4	-	1
<i>Develop bot</i>	6	8	-	3
PC8. Identify the objects for bot development from the Technical Design Document	2	3	-	1
PC9. Identify the tools for developing bot as per the design documents	2	3	-	1
PC10. Generate the software code or bots for each component of Robotic Process Automation solution	2	2	-	1
<i>Integrate</i>	2	2	-	1
PC11. Integrate various modules of the bot or Robotic Process Automation solution	2	2	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Best Practices</i>	10	10	-	5
PC12. Develop modular codes or programs so that they can be reused	2	2	-	1
PC13. Use various techniques, such as use of configuration file, to avoid hardcoding external settings (file paths, URLs, etc.)	2	2	-	1
PC14. Apply code quality principles and best practices, such as secure coding practices naming convention standards and compliance, zero usage of junk code, componentization etc. to maintain readability of the code	2	2	-	1
PC15. Align the code development process to Agile methodology principles to coordinate with cross-functional teams, and ensure timely delivery of the bots	2	2	-	1
PC16. Ensure compliance with internal control or audit requirements	2	2	-	1
<i>Documentation</i>	2	3	-	1
PC17. Document the software code for consumption and feedback from relevant stakeholders, and include pertinent information such as screenshots, possible exceptions, error handling, etc.	2	3	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8614
NOS Name	Design & Develop the bot as per design documents
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Robotic Process Automation
NSQF Level	5.5
Credits	3
Version	3.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

SSC/N8615: Develop test cases for the bot

Description

This unit is about developing test cases to test the bot

Scope

The scope covers the following :

- Define Requirements
- Create Test Cases
- Run Test Cases
- Documentation

Elements and Performance Criteria

Define Requirements

To be competent, the user/individual on the job must be able to:

- PC1.** Define the type of testing and testing requirements for the bot developed, such as functional testing (unit, integration, etc), non-functional testing (performance, endurance, load, volume, scalability, etc)

Create Test Cases

To be competent, the user/individual on the job must be able to:

- PC2.** Access reusable scenarios, test cases, scripts and tools from the organization's knowledge base
- PC3.** Create or modify software test cases relevant to the requirements
- PC4.** Identify test cases that can be automated, and create or modify scripts for the same
- PC5.** Access or create test data relevant to the requirements

Run Test Cases

To be competent, the user/individual on the job must be able to:

- PC6.** Run the simulated test cases and evaluate the outcomes
- PC7.** Communicate the outcomes of the tests or simulations with appropriate people and iterate

Documentation

To be competent, the user/individual on the job must be able to:

- PC8.** Create documentation on the tests or simulations for appropriate people

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Organizational policies, procedures and priorities for developing tests or simulations for software systems
- KU2.** Organization's knowledge base and how to access and update information on scenarios, test cases, scripts and tools

Qualification Pack

- KU3.** Who to involve when developing tests or simulations for software systems
- KU4.** Organizational policies and procedures for documenting simulations
- KU5.** The application and use of test or simulation tools
- KU6.** The approval process for testing the bot
- KU7.** Understanding of processes and workflows pertaining to the Robotic Process Automation solution
- KU8.** Different sources of information available for designing tests
- KU9.** Different types of testing such as unit, sub-system, system, automated
- KU10.** Testing requirements issues and their solutions
- KU11.** The principles of designing tests
- KU12.** Different simulation tool providers and their tools
- KU13.** Evaluation and selection of the best testing tool for the given purpose
- KU14.** Test automation tools
- KU15.** Creation of suitable tests for particular applications
- KU16.** Check whether test cases are suitable to be automated
- KU17.** Access, create and modify different types of: i. test cases, ii. automatic scripts, iii. test data, iv. test plans
- KU18.** Procedures to check the tests are working and fit for the purpose
- KU19.** Analyze and use feedback to improve test cases
- KU20.** Scripting / programming languages to understand test cases
- KU21.** Identify and refer anomalies
- KU22.** Various operating systems such as Linux, Ubuntu, or Windows

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Follow instructions, guidelines, procedures, rules and service level agreements
- GS2.** Work independently as well as collaboratively
- GS3.** Communicate with others in writing
- GS4.** Understand business impact and disseminate relevant information to others
- GS5.** Analyse exceptions and understand its implications on business
- GS6.** Seek, receive and act on gathered feedback from various sources
- GS7.** Share the assimilated data and feedback with stakeholders
- GS8.** Apply good attention to detail
- GS9.** Ensure the work is complete and free from errors
- GS10.** Follow rule-based decision-making processes
- GS11.** Plan and organize the work to achieve targets and deadlines
- GS12.** Problem-solving approaches in different situations
- GS13.** Refer anomalies to appropriate stakeholders

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Define Requirements</i>	4	7	-	3
PC1. Define the type of testing and testing requirements for the bot developed, such as functional testing (unit, integration, etc), non-functional testing (performance, endurance, load, volume, scalability, etc)	4	7	-	3
<i>Create Test Cases</i>	16	25	-	11
PC2. Access reusable scenarios, test cases, scripts and tools from the organization's knowledge base	4	7	-	3
PC3. Create or modify software test cases relevant to the requirements	4	6	-	3
PC4. Identify test cases that can be automated, and create or modify scripts for the same	4	6	-	3
PC5. Access or create test data relevant to the requirements	4	6	-	2
<i>Run Test Cases</i>	7	12	-	4
PC6. Run the simulated test cases and evaluate the outcomes	4	6	-	2
PC7. Communicate the outcomes of the tests or simulations with appropriate people and iterate	3	6	-	2
<i>Documentation</i>	3	6	-	2
PC8. Create documentation on the tests or simulations for appropriate people	3	6	-	2
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8615
NOS Name	Develop test cases for the bot
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Robotic Process Automation
NSQF Level	5.5
Credits	2
Version	3.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

SSC/N8616: Identify and fix bugs in the bot

Description

This unit is about identifying and fixing the bugs in the bot

Scope

The scope covers the following :

- Record
- Debug
- Fix
- Log Analysis

Elements and Performance Criteria

Record

To be competent, the user/individual on the job must be able to:

PC1. Record the bug or enter it in the case tracking system

PC2. Identify what the user was doing, what they were expecting and what happened instead

PC3. Analyze if the error is related to the tool, the bot or the target application

Debug

To be competent, the user/individual on the job must be able to:

PC4. Debug the code using debuggers, binary search, or other suitable methods

Fix

To be competent, the user/individual on the job must be able to:

PC5. Identify the line of code where the bug occurs

PC6. Identify the bug type (e.g., unexpected null, bad input, off-by-one, buffer overflow, index out-of-range, etc.)

PC7. Fix the error in the code

Log analysis

To be competent, the user/individual on the job must be able to:

PC8. Log all activities and analyze the logs

PC9. Continue the debugging and logging processes until the bot runs smoothly

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Organizational policies, procedures and guidelines which relate to logging and log analysis

KU2. Organizational policies and procedures for sharing data

KU3. Organizational policies and procedures for documenting results of application tests and log analysis

Qualification Pack

- KU4.** Who to involve while identifying bugs in the bot's code
- KU5.** The range of standard templates and tools
- KU6.** Understanding of processes and workflows pertaining to the Robotic Process Automation solution
- KU7.** Methods and tools for debugging
- KU8.** Analyze application log files
- KU9.** Debugging issues and their solutions
- KU10.** Scripting / programming of languages to understand the bot's source code
- KU11.** Identify and record bugs in the code
- KU12.** Analyze and use feedback to fixing the bugs

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Follow instructions, guidelines, procedures, rules and service level agreements
- GS2.** Work independently as well as collaboratively
- GS3.** Communicate with others in writing
- GS4.** Assess business impact of automation
- GS5.** Analyze exceptions and understand its implications on business
- GS6.** Seek, receive and act on gathered feedback from various sources
- GS7.** Share the assimilated data and feedback with stakeholders
- GS8.** Apply good attention to detail
- GS9.** Ensure the work is complete and free from errors
- GS10.** Follow rule-based decision-making processes
- GS11.** Plan and organize the work to achieve targets and deadlines
- GS12.** problem-solving approaches in different situations
- GS13.** Refer anomalies to appropriate stakeholders

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Record</i>	9	15	-	6
PC1. Record the bug or enter it in the case tracking system	3	4	-	2
PC2. Identify what the user was doing, what they were expecting and what happened instead	3	5	-	2
PC3. Analyze if the error is related to the tool, the bot or the target application	3	6	-	2
<i>Debug</i>	4	7	-	3
PC4. Debug the code using debuggers, binary search, or other suitable methods	4	7	-	3
<i>Fix</i>	10	17	-	7
PC5. Identify the line of code where the bug occurs	3	5	-	2
PC6. Identify the bug type (e.g., unexpected null, bad input, off-by-one, buffer overflow, index out-of-range, etc.)	3	5	-	2
PC7. Fix the error in the code	4	7	-	3
<i>Log analysis</i>	7	11	-	4
PC8. Log all activities and analyze the logs	4	7	-	2
PC9. Continue the debugging and logging processes until the bot runs smoothly	3	4	-	2
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8616
NOS Name	Identify and fix bugs in the bot
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Robotic Process Automation
NSQF Level	5.5
Credits	2
Version	3.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

SSC/N8618: Ensure that processes are being triggered as intended and handle process exceptions

Description

This unit is about validating trigger types and exception handling in the RPA solution

Scope

The scope covers the following :

- Validate trigger types
- Identify exceptions
- Document

Elements and Performance Criteria

Validate trigger types

To be competent, the user/individual on the job must be able to:

- PC1.** Validate the trigger types in the Robotic Process Automation solution and identify the triggers not addressed by the code

Identify exceptions

To be competent, the user/individual on the job must be able to:

- PC2.** Ensure that the input data conforms to pre-defined criteria
- PC3.** Identify possible issues with underlying applications such as server crash, slow/ un-responsive bot, changes in application, changing to new environment, etc
- PC4.** Identify non-conformity to expected response times
- PC5.** Identify infrastructural issues, such as overloading
- PC6.** Identify business scenarios that have not been handled during process configuration
- PC7.** Include business cases which might be not conforming to the happy path

Document

To be competent, the user/individual on the job must be able to:

- PC8.** Create an exception log containing relevant data such as error message string, unique identifiers, etc
- PC9.** Communicate the information to the development team and appropriate stakeholders to refine the software code
- PC10.** Create documents for exception-handling, and update the business continuity plan in consultation with appropriate stakeholders
- PC11.** Create documents for negative cases and black boxes

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

Qualification Pack

- KU1.** Organizational policies, procedures and priorities for identifying exceptions and handling them
- KU2.** Organization's knowledge base and how to access and update information on exception handling
- KU3.** Who to involve when validating trigger types, documenting exception handling and updating business continuity plans
- KU4.** Understanding of processes and workflows pertaining to the Robotic Process Automation solution
- KU5.** Different sources of information available for exception handling
- KU6.** Types of errors that can be encountered while running the Robotic Process Automation process
- KU7.** Exception handling methods
- KU8.** Exception handling options in standard Robotic Process Automation tools
- KU9.** Scripting / programming languages to understand source code of different components of the RPA solution
- KU10.** Various operating systems such as Linux, Ubuntu, or Windows

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Make decisions on suitable courses
- GS2.** Problem-solving approaches in different situations
- GS3.** Analyze data and activities
- GS4.** Seek, receive and act on gathered feedback from various sources
- GS5.** Configure data and disseminate relevant information to others
- GS6.** Balanced judgments to different situations
- GS7.** Ensure work is complete and free from errors

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Validate trigger types</i>	3	5	-	2
PC1. Validate the trigger types in the Robotic Process Automation solution and identify the triggers not addressed by the code	3	5	-	2
<i>Identify exceptions</i>	18	30	-	12
PC2. Ensure that the input data conforms to pre-defined criteria	3	5	-	2
PC3. Identify possible issues with underlying applications such as server crash, slow/ un-responsive bot, changes in application, changing to new environment, etc	3	5	-	2
PC4. Identify non-conformity to expected response times	3	5	-	2
PC5. Identify infrastructural issues, such as overloading	3	5	-	2
PC6. Identify business scenarios that have not been handled during process configuration	3	5	-	2
PC7. Include business cases which might be not conforming to the happy path	3	5	-	2
<i>Document</i>	9	15	-	6
PC8. Create an exception log containing relevant data such as error message string, unique identifiers, etc	3	5	-	2
PC9. Communicate the information to the development team and appropriate stakeholders to refine the software code	2	4	-	2
PC10. Create documents for exception-handling, and update the business continuity plan in consultation with appropriate stakeholders	2	3	-	1
PC11. Create documents for negative cases and black boxes	2	3	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8618
NOS Name	Ensure that processes are being triggered as intended and handle process exceptions
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Robotic Process Automation
NSQF Level	5.5
Credits	2
Version	3.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

SSC/N8622: Review post deployment analytics & provide process improvement information to appropriate stakeholders

Description

This unit is about monitoring and evaluating the Robotic Process Automation process post deployment

Scope

The scope covers the following :

- Define KPIs
- Monitor
- Evaluate
- Report

Elements and Performance Criteria

Define KPIs

To be competent, the user/individual on the job must be able to:

- PC1.** Identify the KPIs and BI metrics for monitoring the processes of RPA solution, as defined in the design document

Monitor

To be competent, the user/individual on the job must be able to:

- PC2.** Monitor overall utilization pattern of the RPA process and corresponding IT systems
PC3. Monitor overall utilization pattern of the bot

Evaluate

To be competent, the user/individual on the job must be able to:

- PC4.** Evaluate the ability of the bot to scale based on variations in demand
PC5. Analyze the accuracy of the solution and evaluate compliance deficiencies due to data
PC6. Analyze the performance of the solution based on process velocity (time taken to complete the process)
PC7. Identify the bottlenecks arising out of increased throughput
PC8. Assess the downtime of the bot during the runtime
PC9. Assess and compare estimated vs. actual effort utilized in deploying the Robotic Process Automation solution
PC10. Evaluate other metrics as per the design document

Report

To be competent, the user/individual on the job must be able to:

- PC11.** Report the metrics and evaluation results to appropriate stakeholders
PC12. Perform continuous reviews and provide suggestions on process improvement to appropriate stakeholders

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** Organizational policies, procedures and guidelines which relate to monitoring RPA processes and deployed systems
- KU2.** Organizational policies and procedures for sharing data
- KU3.** Organizational policies and procedures for documenting performance insights of the deployed RPA solution
- KU4.** Who to involve for identifying KPIs and sharing reports on performance of the deployed solution
- KU5.** The range of standard templates and tools available for performance monitoring
- KU6.** Understanding of processes and workflows pertaining to the Robotic Process Automation solution
- KU7.** Monitor and optimize utilization of resources
- KU8.** Methods to evaluate scalability, accuracy, process velocity, and other parameters mentioned in design documents
- KU9.** Define metrics to monitor deployed systems
- KU10.** Different types of monitoring tools
- KU11.** Interpret results of performance metrics

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Seek clarification and advice from appropriate people
- GS2.** Listen effectively and orally communicate information accurately
- GS3.** Work independently as well as collaboratively
- GS4.** Communicate with others in writing
- GS5.** Understand Robotic Process Automation process related decisions on business and organization
- GS6.** Seek, receive and act on gathered feedback from various sources
- GS7.** Share the assimilated data and feedback with stakeholders
- GS8.** Ensure the work is complete and free from errors
- GS9.** Plan and organize your own work to achieve targets and deadlines

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Define KPIs</i>	3	5	-	2
PC1. Identify the KPIs and BI metrics for monitoring the processes of RPA solution, as defined in the design document	3	5	-	2
<i>Monitor</i>	6	10	-	4
PC2. Monitor overall utilization pattern of the RPA process and corresponding IT systems	3	5	-	2
PC3. Monitor overall utilization pattern of the bot	3	5	-	2
<i>Evaluate</i>	19	31	-	12
PC4. Evaluate the ability of the bot to scale based on variations in demand	3	5	-	2
PC5. Analyze the accuracy of the solution and evaluate compliance deficiencies due to data	3	5	-	2
PC6. Analyze the performance of the solution based on process velocity (time taken to complete the process)	3	5	-	2
PC7. Identify the bottlenecks arising out of increased throughput	3	4	-	2
PC8. Assess the downtime of the bot during the runtime	3	4	-	2
PC9. Assess and compare estimated vs. actual effort utilized in deploying the Robotic Process Automation solution	2	4	-	1
PC10. Evaluate other metrics as per the design document	2	4	-	1
<i>Report</i>	2	4	-	2
PC11. Report the metrics and evaluation results to appropriate stakeholders	1	2	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Perform continuous reviews and provide suggestions on process improvement to appropriate stakeholders	1	2	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8622
NOS Name	Review post deployment analytics & provide process improvement information to appropriate stakeholders
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Robotic Process Automation
NSQF Level	5.5
Credits	1
Version	3.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

SSC/N8629: New Technology application in RPA

Description

This NOS describes the New technology application RPA

Scope

The scope covers the following :

- New technology application

Elements and Performance Criteria

New technology application

To be competent, the user/individual on the job must be able to:

- PC1.** Describe RPA and its importance in streamlining business processes through automation.
- PC2.** Incorporate Machine Learning, Natural Language Processing, and Computer Vision into RPA solutions to boost automation effectiveness.
- PC3.** Create and deploy RPA bots with popular tools while integrating them with emerging technologies.
- PC4.** Utilize cloud-based solutions for RPA, comprehending deployment approaches and the advantages of cloud resources.
- PC5.** Implement best practices for governance and compliance to safeguard data privacy and security in RPA initiatives.
- PC6.** Design and outline workflows for automation, pinpointing appropriate processes for RPA execution.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge and Understanding of RPA
- KU2.** Knowledge and Understanding of Integrating Technologies
- KU3.** Knowledge and Understanding of RPA Development Tool
- KU4.** Knowledge and Understanding of Cloud Computing
- KU5.** Knowledge and Understanding of Governance and Compliance (e.g., GDPR, HIPAA)
- KU6.** Knowledge and Understanding of Workflow Design

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Innovative thinking to devise solutions that integrate multiple technologies effectively.
- GS2.** Ability to analyze business processes and identify automation opportunities.

Qualification Pack

- GS3.** Skills in planning, organizing, and executing automation projects.
- GS4.** Strong verbal and written communication abilities for articulating project goals and technical concepts.
- GS5.** Vigilance in maintaining compliance with governance and data security measures.
- GS6.** Critical evaluation of new tools and technologies for potential integration with RPA initiatives.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>New technology application</i>	30	50	-	20
PC1. Describe RPA and its importance in streamlining business processes through automation.	5	9	-	4
PC2. Incorporate Machine Learning, Natural Language Processing, and Computer Vision into RPA solutions to boost automation effectiveness.	5	9	-	4
PC3. Create and deploy RPA bots with popular tools while integrating them with emerging technologies.	5	8	-	3
PC4. Utilize cloud-based solutions for RPA, comprehending deployment approaches and the advantages of cloud resources.	5	8	-	3
PC5. Implement best practices for governance and compliance to safeguard data privacy and security in RPA initiatives.	5	8	-	3
PC6. Design and outline workflows for automation, pinpointing appropriate processes for RPA execution.	5	8	-	3
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8629
NOS Name	New Technology application in RPA
Sector	IT-ITeS
Sub-Sector	
Occupation	Robotic Process Automation
NSQF Level	5.5
Credits	1
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

The assessment will consist of a blend of hands-on practical evaluations, viva-voce, and online proctored scenario-based multiple-choice questions ensuring a thorough evaluation of the individual's proficiency in learning outcomes, practical understanding, and real-world application of concepts.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N8601. Identify and analyse business processes suitable for automation	30	50	0	20	100	10
SSC/N8603. Establish a functional design of the automation solution	30	50	0	20	100	10
SSC/N8607. Define infrastructure requirements for the Robotic Process Automation solution	30	50	0	20	100	10
SSC/N8614. Design & Develop the bot as per design documents	30	50	0	20	100	10
SSC/N8615. Develop test cases for the bot	30	50	0	20	100	10
SSC/N8616. Identify and fix bugs in the bot	30	50	0	20	100	10
SSC/N8618. Ensure that processes are being triggered as intended and handle process exceptions	30	50	0	20	100	10
SSC/N8622. Review post deployment analytics & provide process improvement information to appropriate stakeholders	30	50	0	20	100	10
SSC/N8629. New Technology application in RPA	30	50	0	20	100	10
DGT/VSQ/N0102. Employability Skills (60 Hours)	20	30	-	-	50	10
Total	290	480	-	180	950	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
IT-ITeS	Information Technology - Information Technology enabled Services
BPM	Business Process Management
BPO	Business Process Outsourcing
KPO	Knowledge Process Outsourcing
LPO	LPO
IPO	Information Process Outsourcing

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
Helpdesk	Helpdesk is an entity to which the customers will report their IT problems. IT Service Helpdesk Attendant is responsible for managing the helpdesk.